

GNSS PSEUDORANGE CORRECTIONS

Assigned: 23 September 2026

Due: 5 October 2026 (Courtesy late submission 7 October)

OBJECTIVES

In this assignment we compute and apply satellite clock, relativity, ionospheric delay, and tropospheric delay corrections to the range calculations completed in HW3 and compare multipath errors for different pseudorange types.

We will work with data from PRN14 on August 19, 2026. PRN14 is assigned to SVN77 – a Block III GPS satellite that transmits C/A, P(Y), L1C, L2 and L5 civil signals.

OVERVIEW

- Date: Wednesday, August 19, 2026 = GPS Week 2432 (or 384), TOW 259200-345600/ DOY 231
- User location: NIST (in Boulder, CO): ECEF coordinates: [-1288398.567 -4721696.932 4078625.350] (m)]
- RINEX File: NIST00USA_R_20262310000_01D_30S_MO.rnx
- Broadcast ephemeris file: brdcDOY0.26n (DOY is the 3-digit day of year)
- Hints – I usually read in the rinex file once and then save the data into a .mat file which I then load to speed things up for subsequent runs. Also, when you are first getting started it can be easier to work with a small subset of the data instead of the entire day.

ASSIGNMENT

1. **Starting point** (10pts): Select all the data for **PRN14** from the RINEX obs file to work with for this assignment. Create a time vector just including the times for which you have an observation for the satellite. Compute the expected range (R) for each of these times using your code from HW3.

Compute and plot the difference between the measured C/A code pseudorange (C1C) and the expected range from HW3 (R in meters) versus time for our reference date. *What should it look like?*

$$dPRO = C1C - R$$

Print out the first and last values of dPRO.

2. **Clock error** (10pts): Plot the satellite clock correction (bsv) in meters from the `eph2pvt` function for each measurement time. *What should it look like?*

$$\text{Compute and plot: } dPR1 = C1C - (R - \text{bsv})$$

Print out the first and last values of dPR1.

3. **Relativity** (10pts): Add code to the `eph2pvt` function to compute the relativistic correction (relsv) (meters) based on the orbital elements or GPS position and velocity based on the ephemeris data. Use the negative sign as given in Lecture to be compatible with the IS-GPS documentation.

Compute and plot the relativistic correction (meters) for each measurement time. *What should it look like?*

$$\text{Compute and plot: } dPR2 = C1C - (R - \text{bsv} - \text{relsv})$$

Print out the first and last values of dPR2.

4. **Troposphere** (15pts): Write a function to compute the simple tropospheric delay model (meters) based on the satellite elevation and an assumed zenith delay for the ground station location. For the NIST ground station you may assume the zenith value $z_d = 2$ m. Your function call should look like:

```
[tropo] = tropomodel(zd, elevation)
```

Compute and plot the tropo correction (meters) for each measurement time. *What should it look like?*

Compute and plot $dPR3 = C1C - (R - bsv - relsv + tropo)$

Print out the first and last values of dPR3.

Optional (up to 3pts): Download and implement the UNB3 model

<https://gge.ext.unb.ca/Resources/unb3m/unb3m.html>

Compute and plot the UNB3 tropo correction on the same graph as the simple model.

Compute and plot $dPR3UNB = C1C - (R - bsv - relsv + tropoUNB)$

Print out the first and last values of dPR3UNB.

5. **Ionosphere** (15pts): For the ionospheric correction, rather than using a model-based correction (like Klobuchar) we will combine GPS pseudorange measurements into an ionosphere free combination.

Write a function to compute the iono-free combination of the pseudoranges reported in the RINEX obs file and also the iono correction based on these two measurements. *To allow your function to be used for any combination of measurements pass into it the carrier frequencies ($f1 = 1575.42e6$, $f2 = ?$, $f5 = ?$) as well as the measurements.* Your function call to work with C1C and C2L should look like:

```
[PRIF12, iono] = ionocorr (C1C, f1, C2L, f2)
```

Or to combine C1C and C5Q signals

```
[PRIF15, iono] = ionocorr (C1C, f1, C5Q, f5)
```

Compute and plot the iono correction (meters) based on **C1C** and **C2L** for each measurement time. *What should it look like?*

Compute and plot $dPR4 = PRIF - (R - bsv - relsv + tropo)$

* Notice here we are changing the measurement, not the model like we did in the previous parts.

Print out the first and last values of dPR4.

Optional (up to 3pts): Compute an alternative dual frequency combination using a combination of carrier phase data on two frequencies (CPIF)

Compute and plot the iono correction for your new combo.

Compute and plot $dCP4alt = CPIF - (R - bsv - relsv + tropo)$

Print out the first and last values of dCP4alt.

6. **Discussion** (20pts): **Once everything is working, plot dPR1, dPR2, dPR3, and dPR4 vs time on the same graph**
Discuss each step of your results, commenting on the magnitude and other characteristics of each error source you corrected. Note any unexpected features of your plots. If the results don't turn out the way you expect, describe what you did to investigate possible problems in your code or in the data.

Optional (up to 2 pts): Discuss the impact of UNB3 and/or carrier phase iono corrections.

7. **Multipath** (20pts): To investigate pseudorange multipath on different types of pseudoranges available for the newest GPS satellites, write a function to compute the multipath observable and code-minus-carrier (Lect15 slide31) for a given combination of code & carrier measurements. The function call should look like this:

```
[MP1, CMC1] = mpath(C1C, L1C, f1, L2W, f2)
```

Compute and plot the CMC results for these four codes: C1C, C2W, C2L, C5Q for all the measurement times.

On a separate subplot, show the corresponding signal to noise measurements (S1C, S2W, S2L, S5Q).

For each code, choose which second signal to use for the iono corrections and plot the MP results.

Briefly describe the results. Which measurements would you choose to use for positioning?